



DEEP-TECH INNOVATION · FOOD SECURITY & SAFETY

SensoPack

Biodegradable smart packaging that actively **protects** fresh shrimp and **senses** its true freshness — starting with India's ₹62,408 crore shrimp-export chain, and built to be affordable where refrigeration cannot reach.

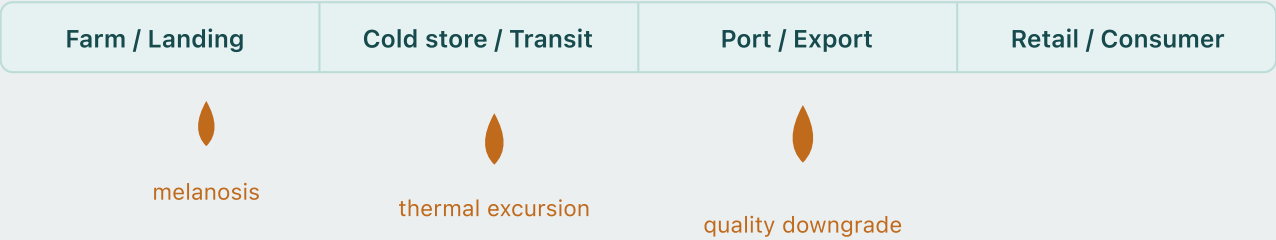
Active bio-film + one low-cost freshness sensor

Beachhead: shrimp · extends to fish & meat

Protect · Sense · Sustain

India exports the world's shrimp — and loses a fortune to spoilage

WHERE THE VALUE LEAKS — POND TO PORT TO PLATE



Shrimp spoils in **3–14 days** by temperature — a static "Best By" label is blind to it

INDIA SHRIMP EXPORT vs LOSS (₹ cr)



69%

of India's seafood dollar earnings come from shrimp

\$2.71B

India → USA shrimp alone (FY25) — the premium market

~35%

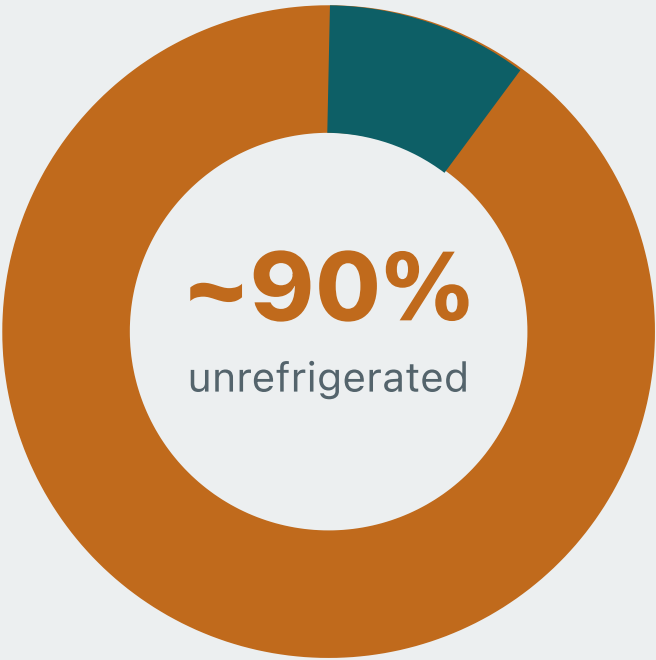
of harvested fish/seafood lost globally before a plate

2 gaps

no active **protection**; no real-time **information**

The fixes exist — they never reach the market that needs them

WHO THE COLD CHAIN COVERS



■ Organised / export (~10%) ■ Domestic / informal (~90%)

FIX 1 — TOO COSTLY

Sensor tags > \$2.50/unit

TrackingPacking-class IoT tags exceed the 5–8% of product value a shrimp pack can bear. Structurally absent from the informal market.

FIX 2 — UNREADABLE

Needs special scanners

Senoptica-type optical labels need industrial readers — a consumer smartphone can't read them at retail.

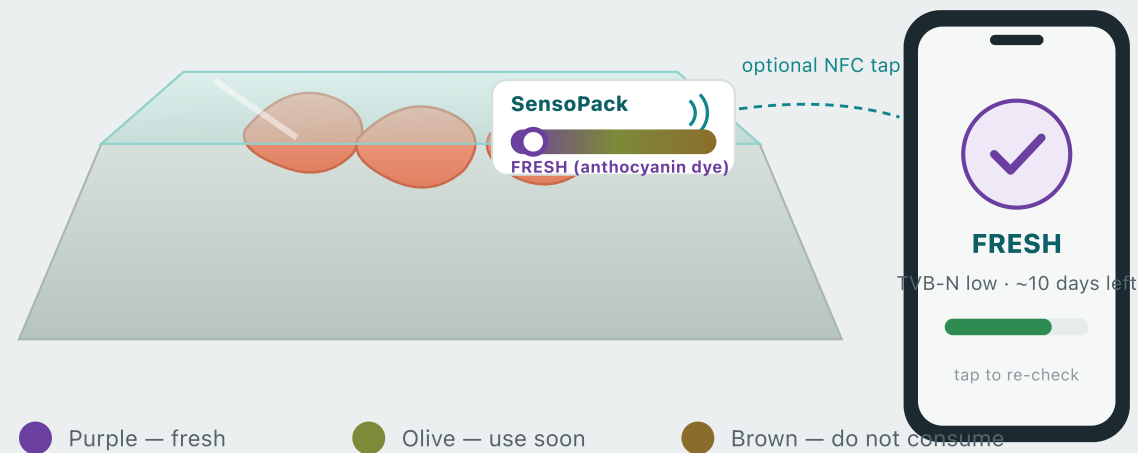
FIX 3 — BOLT-ON

Not packaging-integrated

Blakbear-style add-on tags aren't built into the film — extra steps, extra cost. And "Best By" ignores temperature entirely.

The unserved opportunity: an **affordable**, integrated solution that both **extends** shrimp freshness and **proves** it — readable by the naked eye, deployable with no cold storage at all.

What SensoPack looks like in the customer's hands



Product visualisation (concept render — physical unit in development).

PROTECT

Active biodegradable film

Chitosan + essential oils (thymol/carvacrol/clove) — an active barrier that **doubles shrimp shelf life** and works without refrigeration.

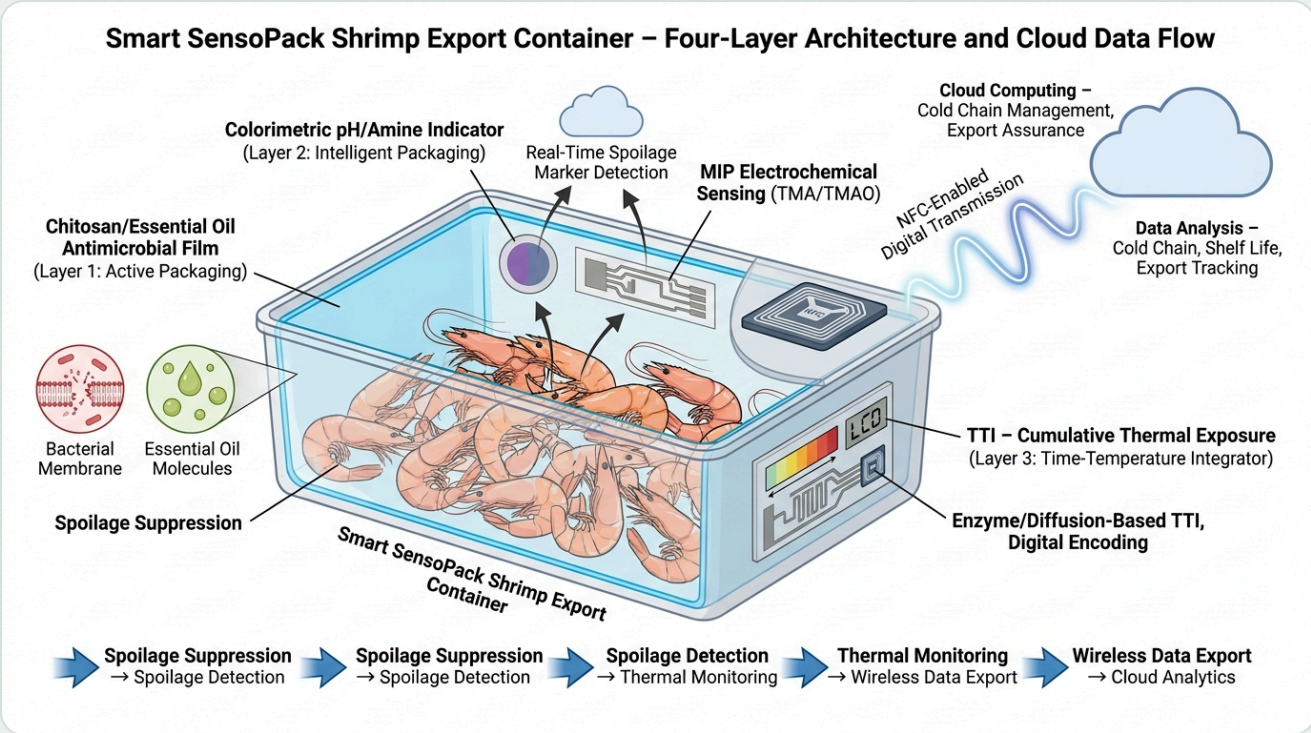
SENSE

One <₹5 colour indicator

An **anthocyanin dye patch** (butterfly pea / roselle) that shifts colour with spoilage amines — **naked-eye readable, no electronics**, optional NFC tap for organised buyers.

One decisive sensor, not a stack of five. Cheaper, more robust, genuinely affordable.

An affordable base — with a full export-tier platform



Full-platform architecture (premium / export tier). The mass-market **base tier** is the film + ₹5 colour patch only.

BASE TIER · MASS MARKET

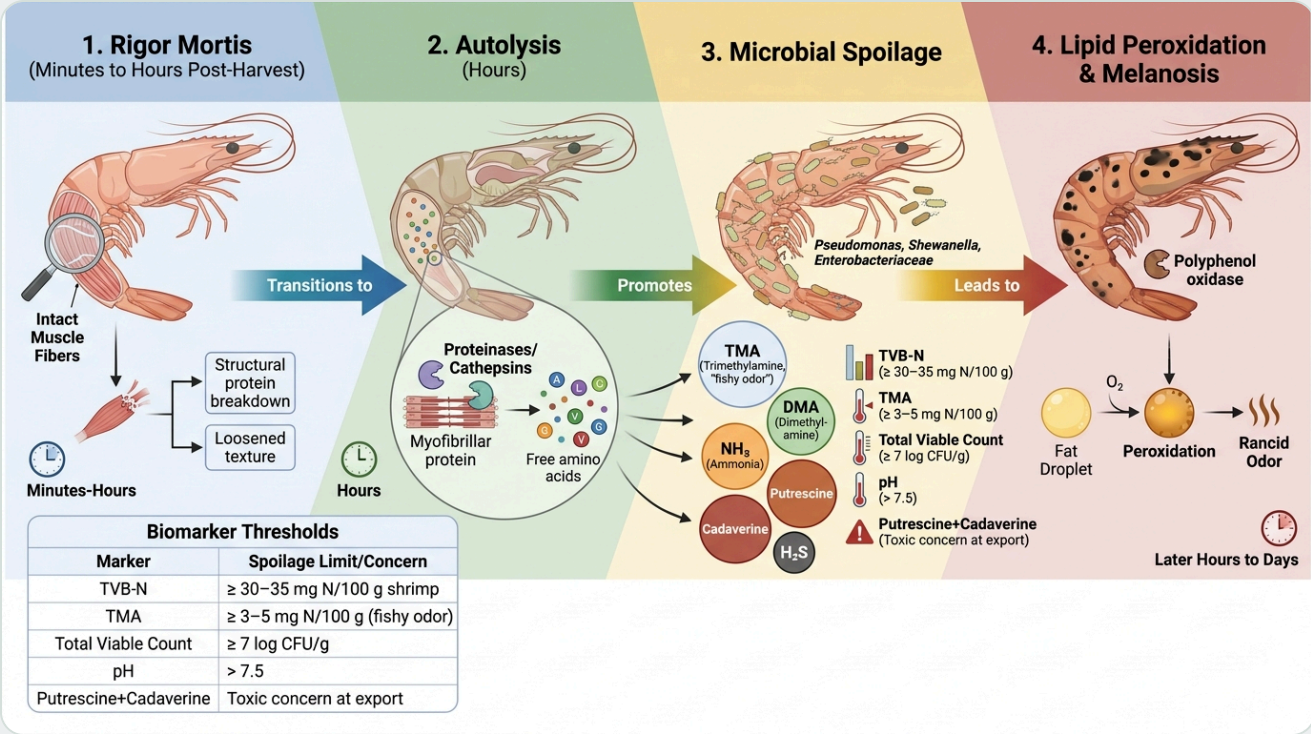
Active **chitosan + EO film** + a **<₹5 anthocyanin colour patch** — naked-eye, no electronics, no battery. What a wet-market vendor actually buys.

PREMIUM / EXPORT TIER · OPTIONAL

Adds electronic amine sensing (MIP), an NFC digital freshness certificate, and a time-temperature log — for MPEDA exporters who need a cold-chain audit trail.

One decisive signal for everyone; electronics only where the value justifies it. We never force a five-sensor stack onto a ₹200 tray.

Measuring the markers regulators already trust



The four-stage shrimp spoilage cascade & accepted biomarker thresholds. Ref: Das & Mishra (2023).

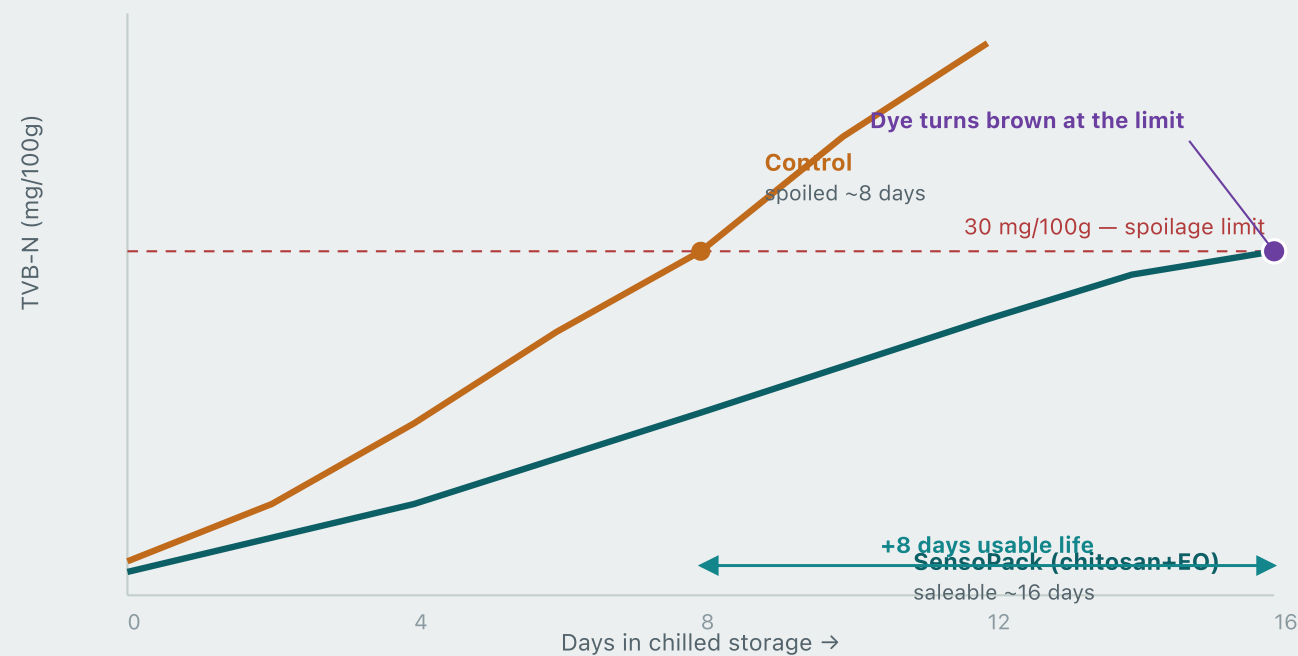
Protect — established science

- Chitosan + EO coatings extend chilled shrimp ~8 → 16 days (published)
- Low OTR arrests oxidation & melanosis; balanced WVTR stops drip loss
- Antimicrobials extend the microbial lag phase (stages 3–4)

Sense — the earliest reliable signal

- We target **stage-3 amines** — TVB-N ≥30, TMA ≥3 mg N/100g
- Anthocyanin dye shifts colour with these amines — **correlated with TVB-N**
- Compensated read-out cancels humidity/temperature drift — the deep-tech edge

Double the shelf life — and flag the exact moment it ends



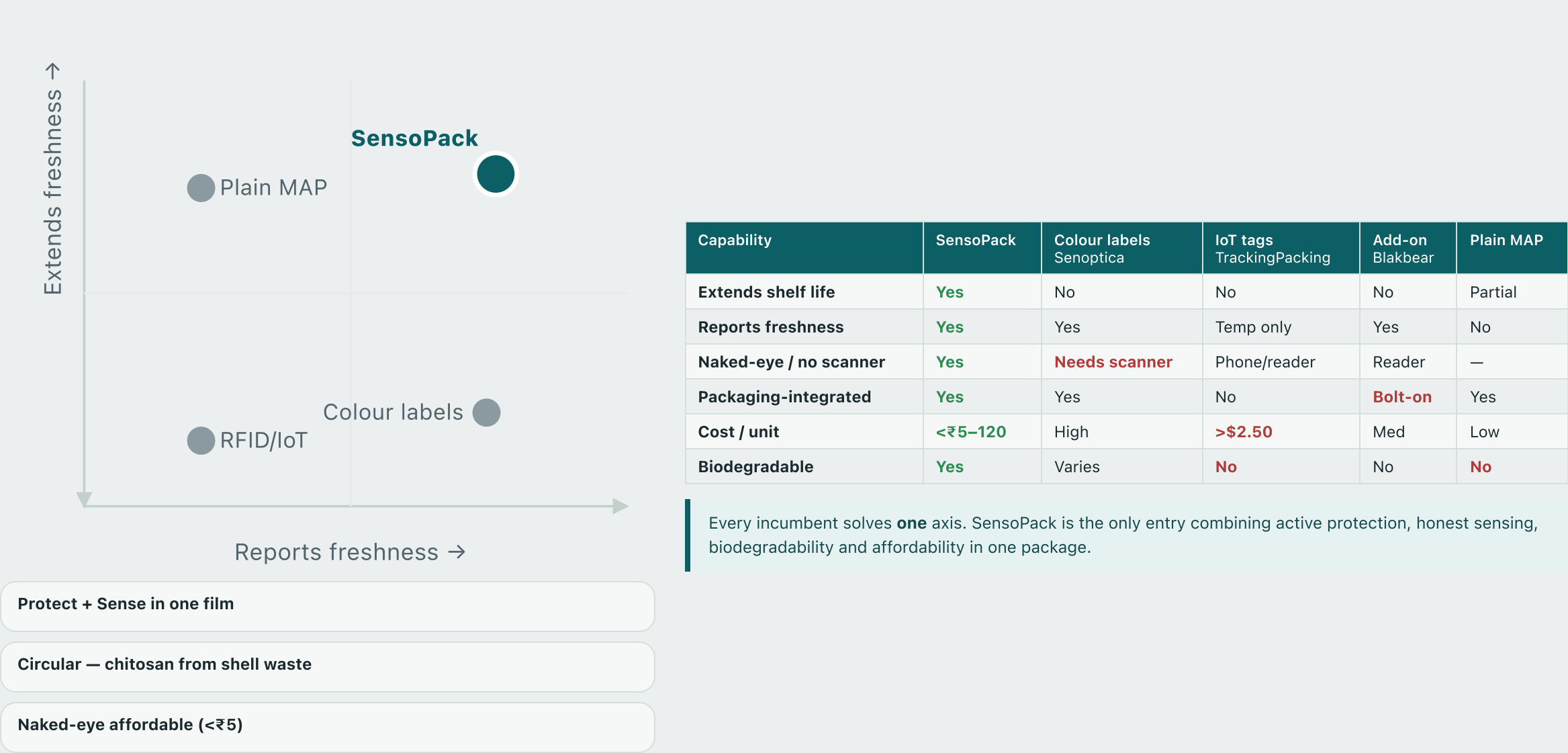
Based on published chitosan-EO shrimp shelf-life data (control ~8 d → ~16 d); our validation trials to confirm on Indian supply.

8 → 16
days of chilled shelf life (published chitosan-EO coatings)

Arrhenius
shelf life: 0°C ≈ 14 d · 5°C ≈ 7 d · 10°C ≈ 3–4 d — a fixed date can't capture this

1 dye
flips at the exact TVB-N limit — no false "fresh"

The only solution that both protects and reports — affordably



From a rural pond to a US hypermarket

SCALE

₹62,408 cr

India shrimp export (FY25); ~₹15,000 cr lost to spoilage/downgrades — a fraction recovered is enormous.

DEPTH

8 → 16 d

doubled shelf life ends distress pricing, displaces carcinogenic formalin/ammonia-ice, preserves drip-loss weight.

REACH

3 tiers

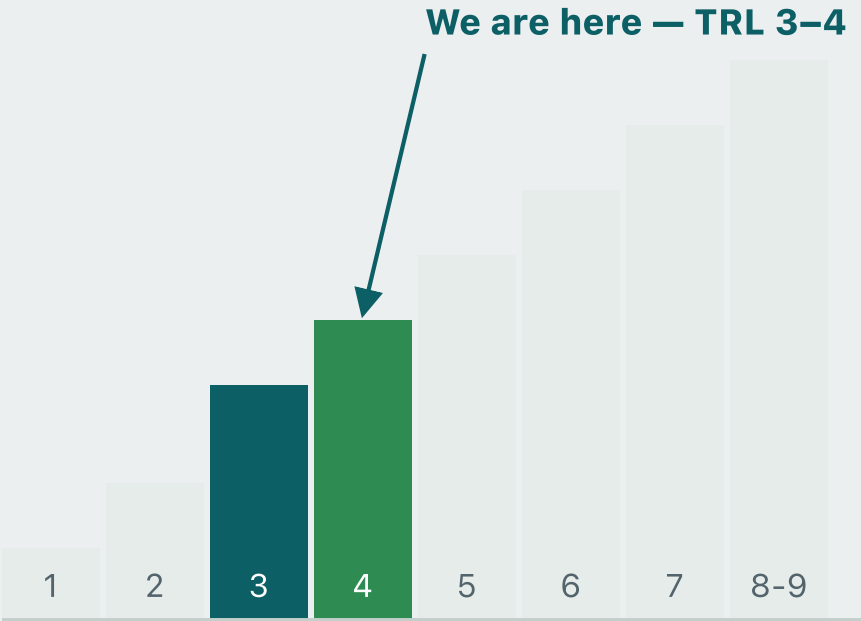
one material serving wet market, organised retail and premium export.



Not a premium gadget for the organised cold chain — **quality & safety infrastructure for the part of the food system that has none.**

A real material, an honestly-stated stage

TECHNOLOGY READINESS LEVEL



Built

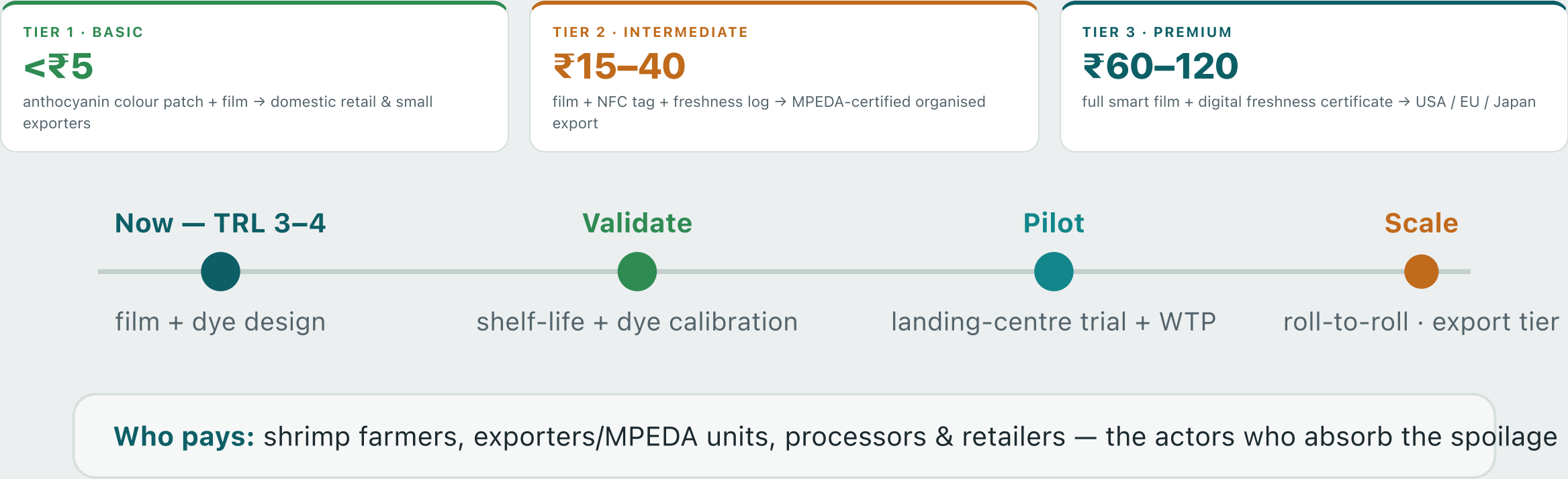
A **cast bio-film prototype** with high optical transparency and coherent, flexible film formation — the two commercially critical properties.

Next (8–12 wks)

Quantitative OTR/WVTR/tensile, antimicrobial assays, anthocyanin-dye calibration to TVB-N, and shrimp shelf-life trials.

Parameter	Prototype status	Scale-up hurdle
Transparency	High clarity, no phase separation	Hold on thin-gauge rolls
Integrity	Clean edges, uniform, no cracking	Survive heat-seal / MAP
Resilience	Well-formed standalone sheet	Barrier under humid cold storage

A cost tier for every buyer — from vendor to exporter



Every judging criterion, answered

Criterion	SensoPack's answer
1 • Problem	India's ₹62,408 cr shrimp export bleeds ~₹15,000 cr to spoilage — scoped to the unrefrigerated segment others ignore.
2 • Originality	The only solution that protects and senses in one biodegradable, circular, naked-eye-affordable package.
3 • Soundness	Chitosan-EO shelf-life (8→16 d) + anthocyanin sensing keyed to accepted TVB-N/TMA thresholds — one sensor, not five.
4 • Impact	Export earnings protected, formalin displaced, waste recovered — across three market tiers.
5 • Prototype	A real cast-film prototype at TRL 3–4 with a concrete validation roadmap — stated honestly.
6 • Viability	<₹5 to ₹120 cost tiers, capital-light, staged from wet market to premium export.



Protect. Sense. Sustain.

SensoPack — affordable, biodegradable smart packaging keeping India's shrimp safe, fresh and saleable. · Team Sentinel · Thank you.